

Romain Giraud

✉ job@romaingiraud.com | romaingiraud.com

🌐 www.linkedin.com/in/romain-giraud-com/ | github.com/romaingiraud

Senior Software Engineer specialized in **C/C++** and **performance-critical systems**, with additional experience in **Python** and **machine learning**. Used to working in R&D contexts, driving software from PoC to production, and delivering highly optimized, reliable systems in telecom and avionics environments, from low-level embedded components to backend and ML-driven services.

Education

Linux Foundation

Certified Kubernetes Application Developer

Online

Dec. 2024

Polytech'Nice

Computer engineering degree

Sophia Antipolis, France

Aug. 2008 – Aug. 2011

Experience

Senior Engineer

FraudBuster

Oct. 2020 – May 2025

Marseille, France

- Designed and implemented a **real-time 4G traffic analysis system** capable of processing high-volume network data, using **C** and **DPDK** for low-latency packet handling and a **Python/Django backend** with **Celery**, **PostgreSQL**, and **MongoDB**, enabling scalable deep packet inspection in a telecom environment.
- Designed and implemented a **proof of concept ML pipeline** to detect fraudulent calls on a GSM network, combining **Go-based network capture and audio processing** with **PyTorch models** to generate decision-support indicators for fraud analysts, after evaluating statistical, speech-recognition, spectrogram-based, and deep learning approaches.
- Joined a newly formed development team following a company merger, contributing to the design of a Mobile Money product by **building data collection and transformation tools for mobile money transactions**, enabling analysts to efficiently monitor and investigate financial activity.
- Trained and deployed **machine learning models** to classify scam messages in a Mobile Money fraud context, providing **decision-support indicators** to assist fraud analysts.

Paternity leave

Jan. 2020 – Sept. 2020

World travel

Sept. 2018 – June 2019

Lead Engineer

Nex Vision

June 2014 – June 2018

Marseille, France

- Designed and implemented an embedded video surveillance system for the Airbus A380, enabling real-time display of multiple camera feeds on a single screen, managing video data flow and integration with the Cabin Management System, using GStreamer on a Texas Instruments chipset.
- Designed and architected an embedded gyrostabilized camera system for helicopter applications, combining a custom PCB with an NVIDIA Tegra, real-time FPGA processing, and a C++/Qt/OpenGL software stack, including low-level Linux driver development for motor and camera control.
- Took ownership of software architecture definition, CI implementation, and technical issue resolution, coordinating resources to support multiple development initiatives.

Freelance

Independent

May 2012 – May 2014

Marseille, France

- Co-created a 2D platformer video game, initially developed in XNA and later migrated to Unity, redesigning the architecture to support the engine transition. Implemented interactive environment systems (water and lava simulation), object-based interactions, and item properties (e.g., wood, stone) to enable gameplay mechanics and puzzle-solving.

Lead Engineer

So Touch

Nov. 2011 – April 2012

Marseille, France

- Designed and implemented an interactive Unity showroom application to showcase search engine capabilities, enabling users to explore travel destinations on a 3D globe. Integrated Kinect, Leap Motion, and 3D display support, including a custom UI system optimized for gesture-based interaction.

Skills

Languages: C++, Python, C, Rust, Go, C#, SQL

Frameworks: Boost, PyTorch, Django

Developer Tools: Git, jj, container (Docker, Podman), CI/CD, Kubernetes

Soft: collaboration and communication skills, curious, fast learner, French native, English fluent